Night shift — full-development SHD comparison

ar068 · 18 July 2026 · Draft

Digest

Empty until the registered experiment finishes.

Mandate

Purpose and scope

This one-seed exploratory shift asks whether PING's modest point advantage in `exp066` becomes clearer when both architectures train on substantially more Spiking Heidelberg Digits (SHD) data under a leakage-resistant train/validation/test protocol. It is intended to identify a claim worth defending later, not to provide multi-seed confirmatory evidence.

The primary question is whether PING outperforms matched COBA in overall official-test accuracy. Accuracy relative to excitatory activity is reported transparently as paired accuracy and firing-rate measurements; no post-hoc composite score will be invented.

Registered experiment

Field	**exp068**
Hypothesis	With matched training and readout settings on substantially more SHD development data, PING exceeds COBA by at least two percentage points in overall official-test accuracy while using less excitatory spike activity.
Baseline	exp066 established one-seed feasibility on 1,000 training utterances: COBA reached 31.0% best held-out accuracy and PING 34.7%. The present contemporaneous matched COBA cell is the architectural baseline; the 5% SHD chance floor remains a diagnostic reference.
Kill criterion	The accuracy hypothesis is falsified if PING does not exceed COBA on overall official-test accuracy. Kill either cell for non-finite loss or logits, persistent skipped updates, silence, clear saturation, or failure to complete the fixed protocol; kill the comparison for different partitions, mismatched registered settings, or any use of official-test data before both checkpoints are frozen.
Seed count	1
Budget	One local plumbing smoke plus one 40-epoch full run per architecture; two concurrent pods maximum, 3 h target per pod, and 10 USD hard total RunPod spend.
Status	queued

Data partition and sealed evaluation

Use SHD's official 8,156-utterance training split and 2,264-utterance test split. At seed 42, partition the official training split deterministically into approximately 90% development training and 10% validation, stratified jointly by class and speaker. COBA and PING receive byte-identical index sets, preprocessing, and sample-ordering policy.

The complete official test split remains sealed during implementation, debugging, training, hyperparameter selection, stopping, and checkpoint selection. After both training runs have finished and both checkpoint choices are frozen, evaluate each selected checkpoint exactly once on the full official test set. Neither result may be revealed or acted upon before both checkpoints are frozen. Overall official-test accuracy is primary. Macro class accuracy and accuracy for seen versus unseen speakers are registered diagnostics; SHD test speakers 4 and 5 form the unseen-speaker group.

Matched architectures and optimisation

Both cells use 256 excitatory and 64 inhibitory cells, Dale-constrained fixed recurrent weights, input-weight mean 0.9, 95% input sparsity, membrane-mean readout with scale 225, Adam learning rate 0.0004, batch size 32, 1 ms integration timestep, 1,000 ms simulation duration, and no firing-rate regulariser. Initialisation, data, preprocessing, ordering, input weights, readout, and every other applicable setting remain matched.

COBA disables the inhibitory loop and uses no voltage-gradient dampening. PING enables the fixed inhibitory loop and uses voltage-gradient dampening
1000. No architecture-specific tuning or rescue is permitted after either

result is seen.

Training and checkpoint selection

First run a small local staged train/validation smoke test to verify the partition, data plumbing, tensor shapes, finite optimisation, checkpointing, and active populations. Smoke measurements are not scientific evidence.

Each full cell then trains for exactly 40 epochs. Select the checkpoint with highest validation accuracy; break equal-accuracy ties with lower validation cross-entropy and then the earlier epoch. The experiment runner must preserve this rule explicitly rather than relying on an incompatible default. Only after both checkpoint identities are recorded may the sealed official-test evaluation begin.

Outcomes and interpretation

The primary exploratory result is δ_{accuracy} = PING accuracy − COBA accuracy on the complete official test split.

1. If the difference is at least +2 percentage points, the result identifies a promising PING accuracy claim for a later confirmatory study.
2. A positive difference below +2 points is a weak directional signal, not a useful accuracy claim.
3. A zero or negative difference provides no PING accuracy advantage under this recipe.

Registered secondary diagnostics are official-test cross-entropy, validation accuracy and loss curves, macro class accuracy, seen- and unseen-speaker accuracy, excitatory and inhibitory firing rates, accuracy alongside excitatory spike rate, runtime, peak GPU memory, non-finite batches, skipped updates, and directly matched input/E/I rasters for preselected official-test utterances.

Integrity and stopping rules
Kill and record a cell if loss or logits become non-finite, skipped updates persist, activity is silent or clearly saturated, or the fixed protocol cannot complete. Kill the comparison if the cells receive different partitions or any registered setting differs. Failed and killed attempts are retained rather than overwritten. The question, baseline, primary outcome, interpretation bands, and kill criteria cannot change after results are observed.

Operating contract

```
budgets:
  wall_clock_target_per_pod: 3h
  wall_clock_per_shift: 8h
  runpod_total_usd: 10
  max_pods_concurrent: 2
  seeds_default: 1
scope:
  collection: spiking-heidelberg-digits
  may_propose: true
  may_edit_snn_cli: false
  experiment_id: exp068
  activity_trace: ar069
stop_when:
  - queue_empty
  - shift_budget_exhausted
  - runpod_budget_exhausted
  - protocol_integrity_failure
  - build_red_twice
```

Two concurrent RunPod pods are authorized: one for COBA and one for PING, preferably using RTX 5090 or an equivalently suitable GPU. Total experiment spend may not exceed 10 USD. Both pods must be reaped immediately after collection or failure. Stop for additional authority before exceeding the budget or pod limit, editing `tools/snn`, changing this design, or taking any other action outside the approved scope.

Implementation should use `experiments/exp068.py` and existing CLI capabilities. Runner-controlled staging of deterministic SHD train/validation files is permitted when provenance is retained and the untouched official test split is used only for final evaluation. Use focused lightweight checks and `demolab build`; do not run the full test suite on the 4 GB host.

Record
Empty until the mandate is approved, committed on `main`, and execution begins.